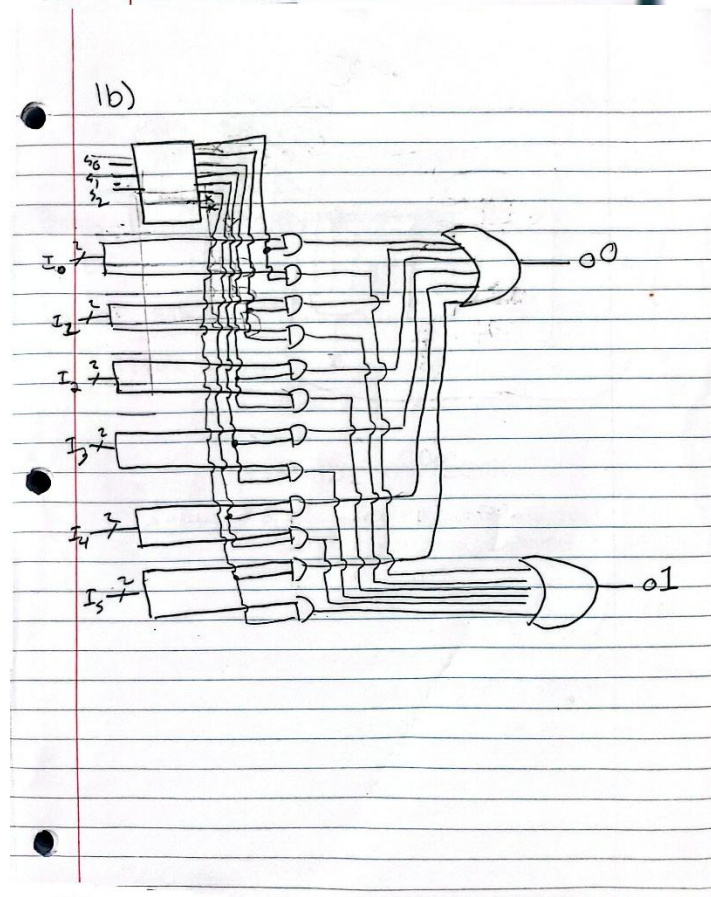
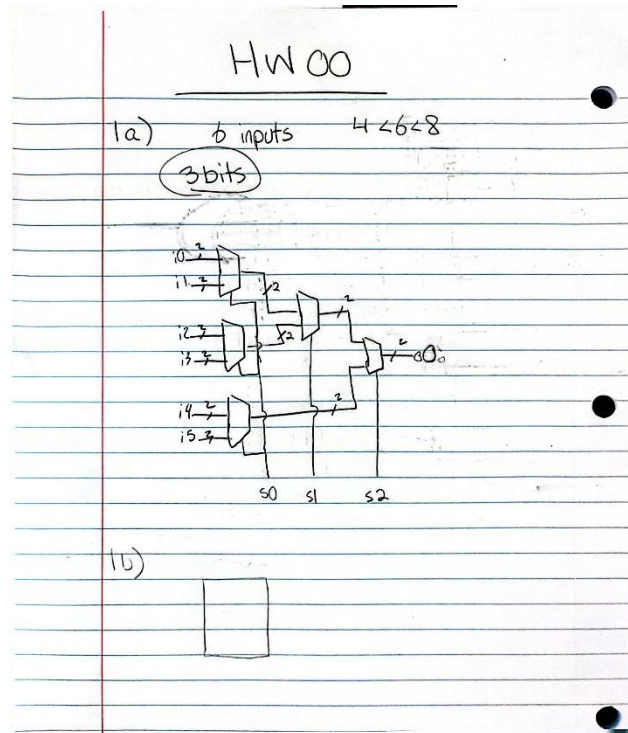
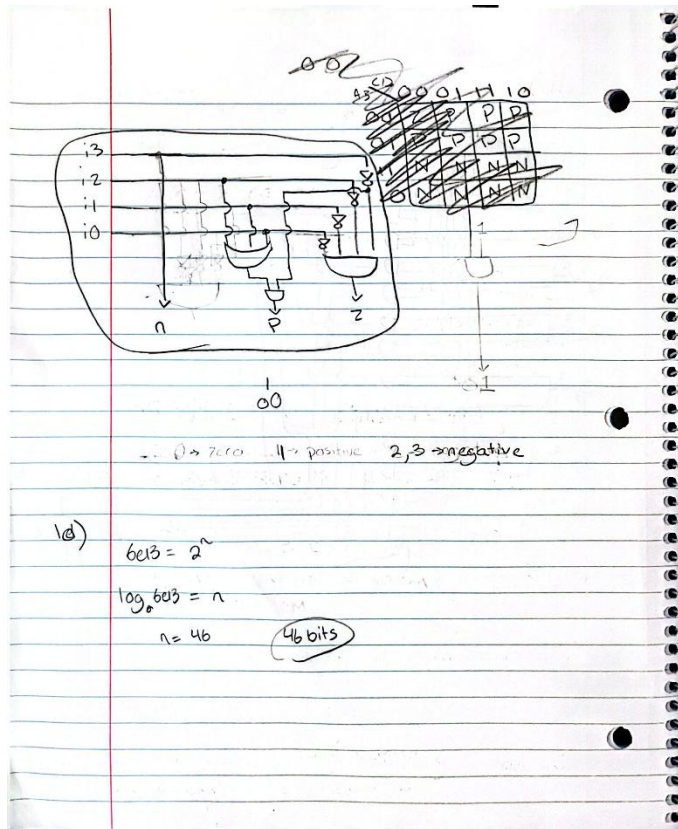


CprE 3810 Hw1

Justin Adrowski

1.





1d) $603 = 2^n$
 $\log_2 603 = n$
 $n = 46$ (46 bits)

1e) $0x38157EE9$ Hex → Binary: each digit = 4 digit Binary

Binary: 0b0011100000010101011111011101001

Decimal: $3 \cdot 16^7 + 8 \cdot 16^6 + 16^5 + 5 \cdot 16^4 + 7 \cdot 16^3 + 14 \cdot 16^2 + 14 \cdot 16 + 9$

= 0d940932841

2. A:

2) 3 Components of Computer

Input

Output

Memory

Control

Datapath

My PC: Dell Precision 5570

• Part numbers & specifics obtained thru Task Manager Components & [dell.com/support/manual/en-us/precision-15-5570...](https://dell.com/support/manual/en-us/precision-15-5570)

• 15.6" WWAN
Wireless WIFI Antenna
Cable Kit

→ Input or Output

* Communicates data & instructions
bidirectionally

• Nvidia RTX
A1000 GPU

→ Datapath & Control

* Performs Matrix Computations to
Generate complex images & sends data
signals to display

• Micron 2200 NVMe 1TB
SSD → Memory

• 16.0 GB DDR5 → Memory
RAM

• Intel(R) WiFi 6E AX211 160 MHz
→ Input/Output

* Receives & transmits wireless
data sending to & from the processor

• 12th Gen Intel(R) Core(TM) i7-12700H
→ Control & Datapath

* It's a CPU, which is specified
as a control/datapath mix in Zybooks

• Intel(R) Iris(R) Xe Graphics

* Built-in GPU within the processor
chip → Control / Datapath

* Same reason as the Nvidia GPU

• Realtek ALC3231 Audio Controller → Input/Output

* Captures & Produces Audio translating between
digital data & communicating the digital data w/
the CPU

B:

2b)

1940s: Manchester Baby

computerhistory.org

en.wikipedia.org

- 32-bit word length
- 32 word memory $\rightarrow$ 1024 Bits

1950s: UNIVAC 1 - 1st commercial stored-program

ethw.org Community Cadence

~~Wikipedia~~ 12 KB or 9 KB if used 6-bit characters

$\downarrow$
72000 Bits

1960s: IBM 1401

Harvard University / IEEE.org

16000 characters of memory $\rightarrow$ 8 bits per character

$\rightarrow$ 128000 bits $\rightarrow$ 128 KB

1970s: Apple II

Smithsonian $\rightarrow$ 4-48 KB ram

Wikipedia

1980s: Gameboy wordpress.com

- 8KB working RAM
- 8KB video RAM

1990s: iMacG3 [Apple.fandom.com](https://apple.fandom.com)

- 32MB RAM - 2MB Video RAM
- 4 GB hard drive

2000s: Nintendo Wii [Copetti.org](https://copetti.org)

- 16KB EEPROM - 4KB Mask ROM
- 512MB NAND Flash storage - 96KB SRAM
- 64 MB GDDR3 SDRAM
- 3MB 1T-SRAM

2010s: Xbox 1 [Ign.com](https://ign.com)

- 500 GB HDD
- 8GB system memory

2020s: Macbook Pro [Apple.com](https://apple.com)

- 16-32 GB 1-2TB SSD
- unified memory

C:

```
PS X:\School\Fall 2026\CprE3810\Hw\Hw1> & "X:\School\Fall 2026\CprE3810\Hw\Hw1\findAvg.exe"
Enter the number of elements: 5
Enter the elements: 7
4
8
11
14
The average of the array elements is: 8.80
PS X:\School\Fall 2026\CprE3810\Hw\Hw1> & "X:\School\Fall 2026\CprE3810\Hw\Hw1\findAvg.exe"
Enter the number of elements: 7
Enter the elements: 26
34
24
13
28
17
9
The average of the array elements is: 21.57
PS X:\School\Fall 2026\CprE3810\Hw\Hw1> █
```